UNDER-DASH POWER BOARD: block diagram. Each box is a sheet (double-click it in KiCad).
Power rails +12V, +5V, +3V3 and GND join the sheets by name (power symbols);
every other connection between sheets is a wire here.

**12 V input and protection**

File: input.kicad_sch

**5 V converter (TPS54360, TI reference design)**

File: buck.kicad_sch

**Inter-box cable: DB-25 to the pod**

5V_POD
BRAKE_RPWM
BRAKE_LPWM
BRAKE_EN
MAGNET_EN
BRAKE_CURRENT
RELAY_LEFT
RELAY_RIGHT
RELAY_HAZARD
RELAY_HORN
RELAY_BEAM
CAN_TX
CAN_RX
CAN_STBY
BOX_PRESENT
BRAKE_LIGHT
KILL_SENSE
HX711_DOUT
HX711_SCK

File: cable.kicad_sch

**3.3 V, the pod's 5 V, the LIDAR feed**

5V_POD

**Brake actuator: buffer, BTS7960, current sense, cable magnet**

BRAKE_RPWM
BRAKE_LPWM
BRAKE_EN
MAGNET_EN
BRAKE_CURRENT

File: actuator.kicad_sch

**Signal relays: indicators, hazards, horn, high beam**

RELAY_LEFT
RELAY_RIGHT
RELAY_HAZARD
RELAY_HORN
RELAY_BEAM

File: relays.kicad_sch

**CAN transceiver for OBD-II**

CAN_TX
CAN_RX
CAN_STBY

File: can.kicad_sch

**Presence, brake light, E-stop sense, kill relay, bench**

BOX_PRESENT
BRAKE_LIGHT
KILL_SENSE
HX711_DOUT
HX711_SCK

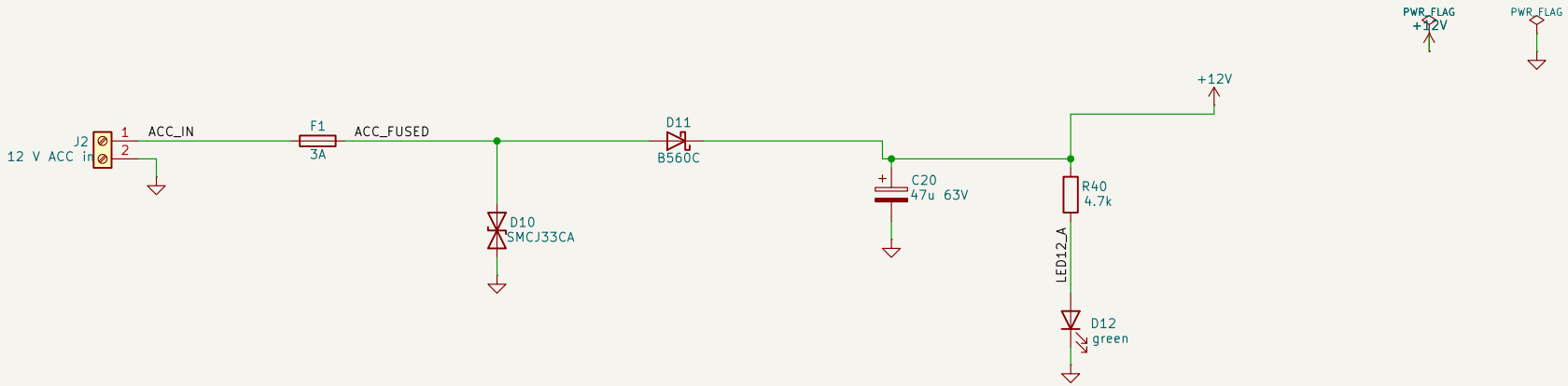
File: status.kicad_sch

12 V input -> 5 V converter -> 3.3 V and the pod's 5 V.
The pod drives the actuator, relays and CAN;
the status sheet reports back to it.

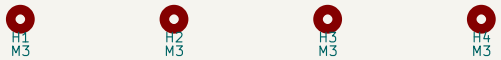
Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Under-dash power board (power, actuator, relays, CAN)
**AR Windscreen FYP - Kiogora Ian Mwenda**

| | | |
|---|---|---|
| Sheet: / | | |
| File: power_board.kicad_sch | | |
| ***Title: Under-dash power board (power, actuator, relays, CAN)*** | | |
| Size: A3 | Date: 2026-10-07 | **Rev: 1** |
| KiCad E.D.A. 10.0.1 | | Id: 1/9 |

12 V INPUT (the logic feed): switched ACC from the car's fuse box, so the board is off with the ignition. F1 3 A fuse. D10 bidirectional TVS clamps load-dump and jump-start spikes to about 39 V. D11 Schottky (60 V, 5 A) blocks a reversed lead. D12: +12V present.




Mounting holes: M3, no electrical connection



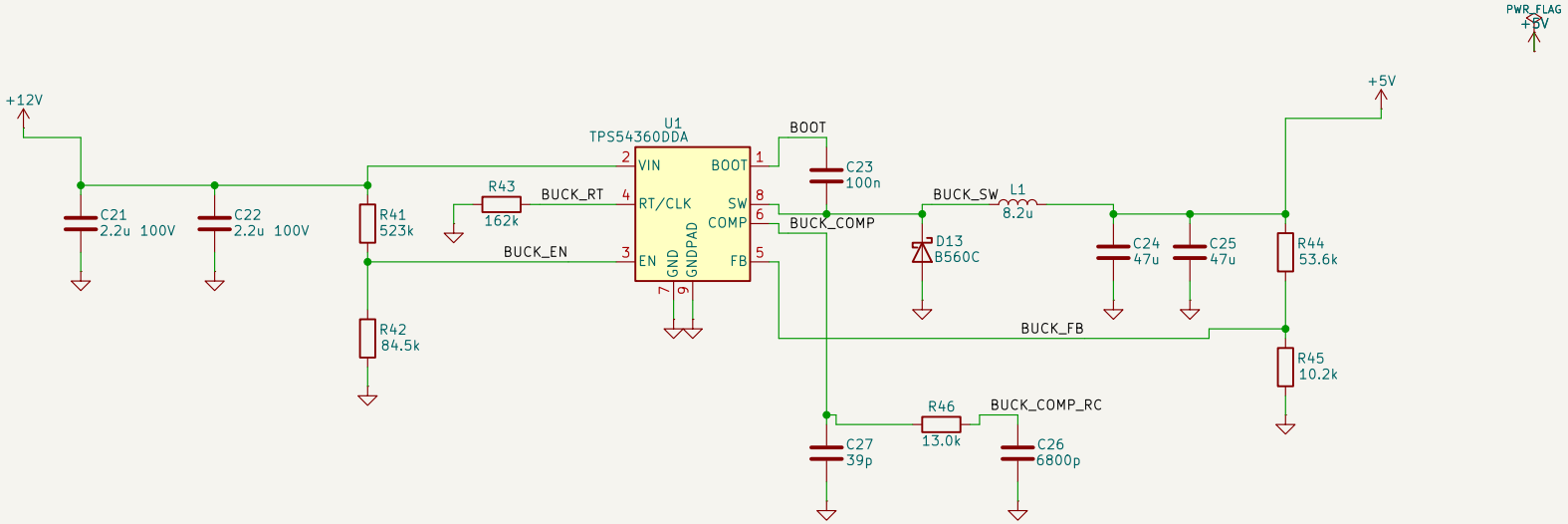

Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Under-dash power board (power, actuator, relays, CAN)
**AR Windscreen FYP — Kiogora Ian Mwenda**

Sheet: /12 V input and protection/
File: input.kicad_sch

***Title: 12 V input and protection***

| Size: A4 | Date: 2026-10-07 | **Rev: 1** |
|---|---|---|
| KiCad E.D.A. 10.0.1 | | Id: 2/9 |

5 V: TI's TPS54360 5 V / 3.5 A reference design (datasheet SLVSBB4G, Figure 34), unchanged.
Rated to 60 V input, so whatever the TVS lets through a load dump cannot destroy it.
R41/R42 start it at 8 V and stop it at 6.25 V. R43 sets 600 kHz. R44/R45 set 5.0 V.
C23 is the bootstrap for the high-side switch; D13 carries the current while it is off.
R46, C26, C27 compensate the control loop. Lay out C21/C22, U1, D13 and L1 as one tight loop.



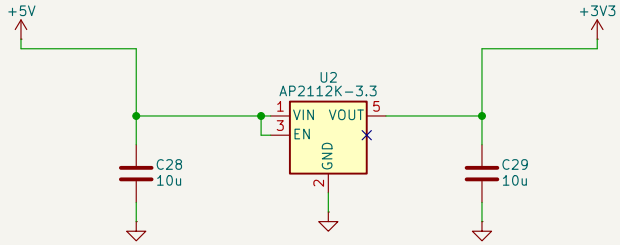

Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Under-dash power board (power, actuator, relays, CAN)
**AR Windscreen FYP — Kiogora Ian Mwenda**

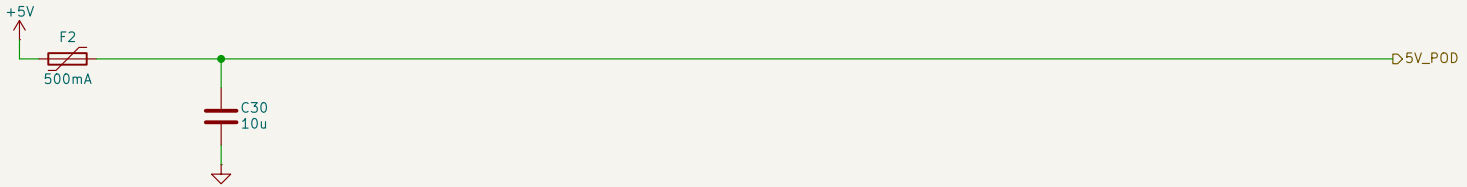
Sheet: /5 V converter (TPS54360, TI reference design)/
File: buck.kicad_sch

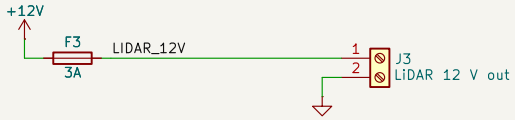
**Title: 5 V converter (TPS54360, TI reference design)**

| Size: A4 | Date: 2026-10-07 | **Rev: 1** |
|---|---|---|
| KiCad E.D.A. 10.0.1 | | Id: 3/9 |

3.3 V (AP2112K) for the CAN transceiver only; the pod makes its own 3.3 V.
5V_POD: the pod's supply up the cable, through F2, a 500 mA resettable fuse,
so a short in the cable cannot take down this board's 5 V.
LiDAR: 12 V through its own 3 A fuse F3 to the M12 socket.



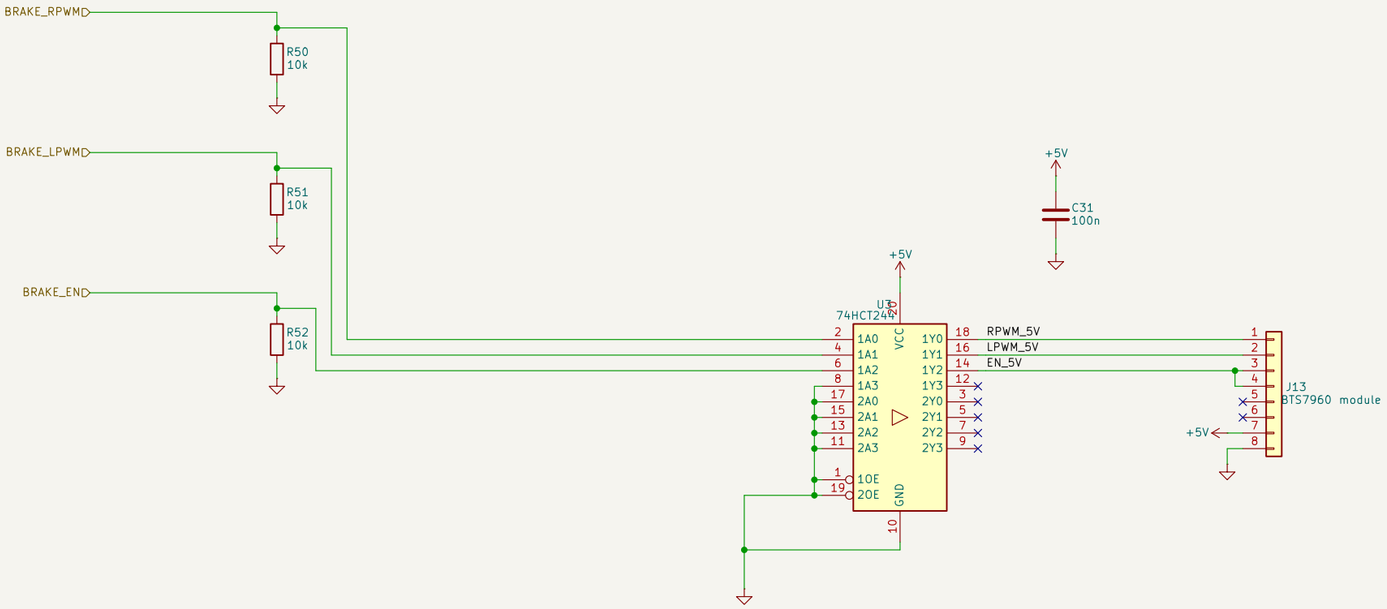

Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Under-dash power board (power, actuator, relays, CAN)
**AR Windscreen FYP — Kiogora Ian Mwenda**

Sheet: /3.3 V, the pod's 5 V, the LiDAR feed/
File: rails.kicad_sch

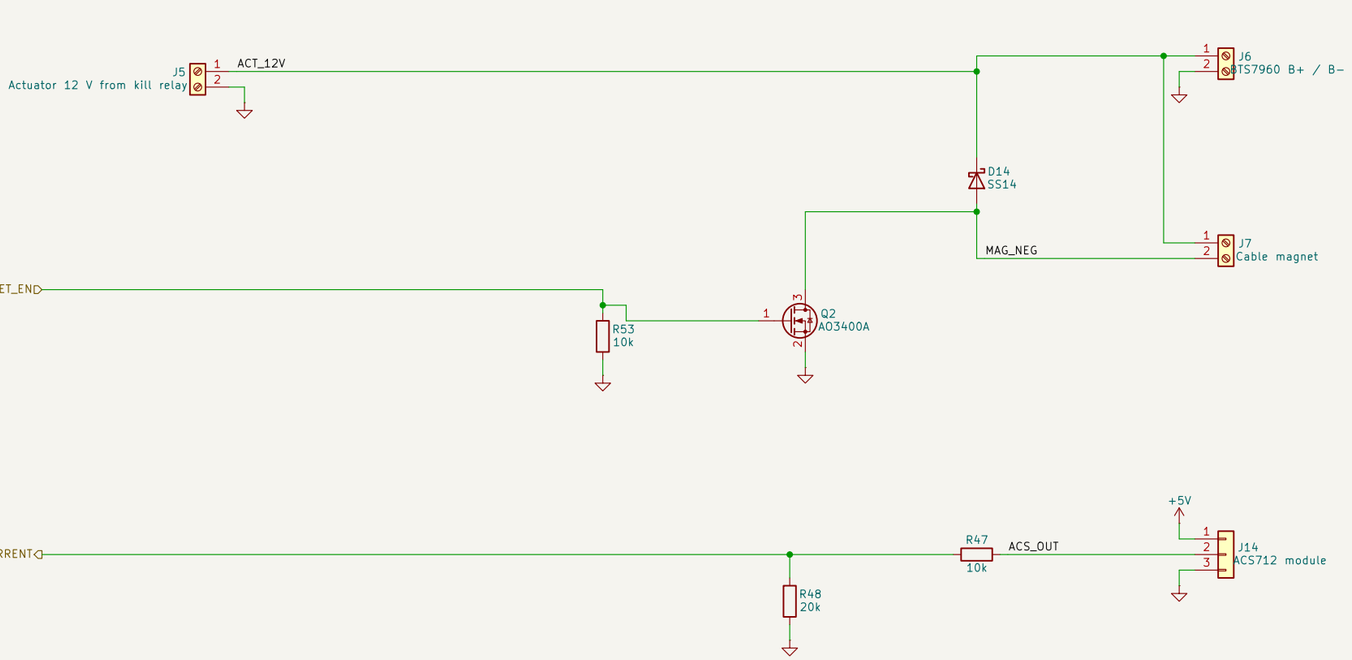
***Title: 3.3 V, the pod's 5 V, the LiDAR feed***

| Size: A4 | Date: 2026-10-07 | **Rev: 1** |
|---|---|---|
| KiCad E.D.A. 10.0.1 | | Id: 4/9 |

BRAKE ACTUATOR. Every line from the pod has a 10 k pull-down here (R50–R53): with the pod in reset, unpowered or unplugged, the drive, enable and magnet lines read LOW: released, cable dropped free. U3 (74HCT244, TTL thresholds) turns the pod's 3.3 V logic into the 5 V the BTS7960 module expects.




Actuator power comes ONLY through the 40 A kill relay's contacts (J5), whose coil is fed through the E-stop. Q2 holds the cable magnet; D14 absorbs its turn-off spike.




ACS712-20A module: 2.5 V at 0 A, 100 mV/A. R47/R48 scale it by 2/3 so nothing above 3.3 V travels up the cable to the pod's ADC.

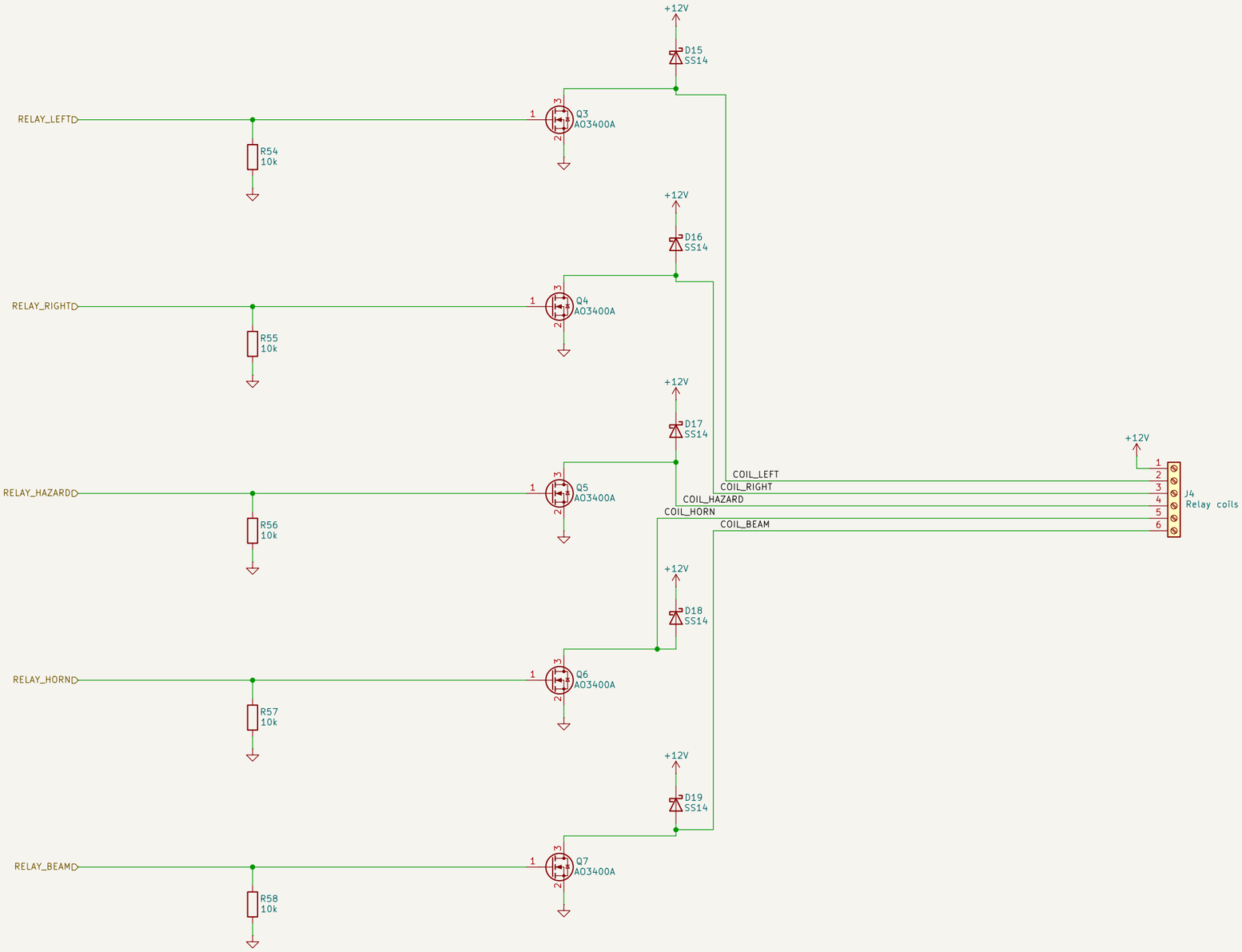
Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Under-dash power board (power, actuator, relays, CAN)
**AR Windscreen FYP — Kiogora Ian Mwenda**

Sheet: /Brake actuator: buffer, BTS7960, current sense, cable magnet/
File: actuator.kicad_sch

***Title: Brake actuator: buffer, BTS7960, current sense, cable magnet***

| Size: A3 | Date: 2026-10-07 | **Rev: 1** |
|---|---|---|
| KiCad E.D.A. 10.0.1 | | Id: 5/9 |

SIGNAL RELAYS: five 12 V automotive relays in sockets, off the board; their contacts tap the car's switch wiring (Part 15.2). Each coil is switched low-side by a MOSFET whose gate has a 10 k pull-down (R54–R58): no drive from the pod, no relay. D15–D19 absorb each coil's turn-off spike.



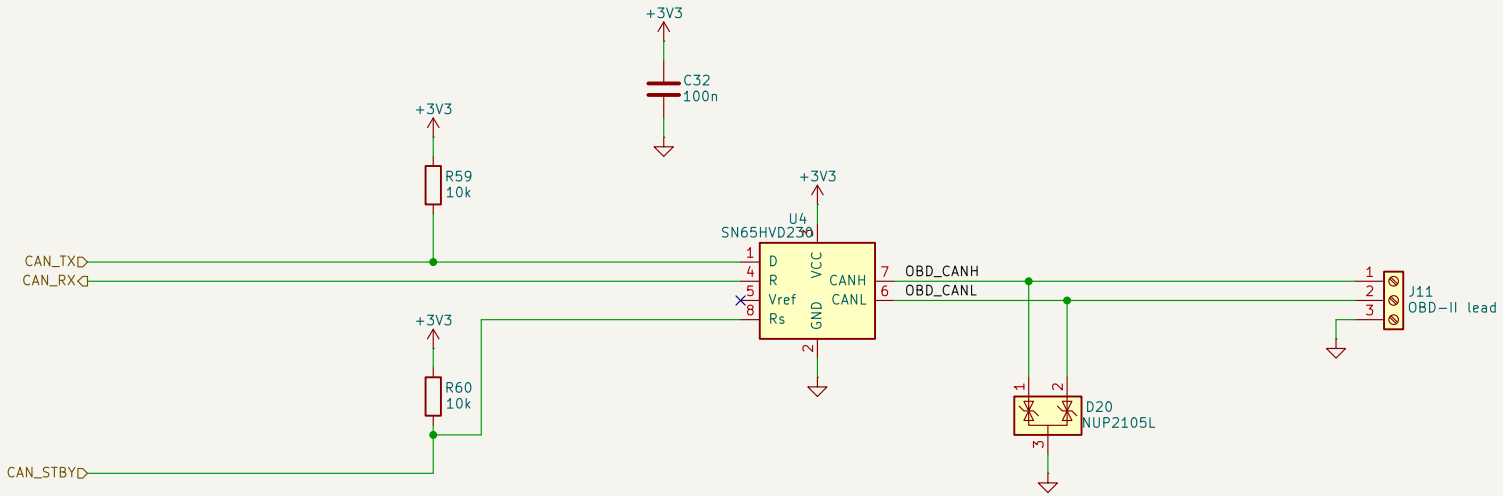

Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Under-dash power board (power, actuator, relays, CAN)
**AR Windscreen FYP — Kiogora Ian Mwenda**

Sheet: /Signal relays: indicators, hazards, horn, high beam/
File: relays.kicad_sch

***Title: Signal relays: indicators, hazards, horn, high beam***

| Size: A3 | Date: 2026-10-07 | **Rev: 1** |
|---|---|---|
| KiCad E.D.A. 10.0.1 | | Id: 6/9 |

CAN (ISO 15765-4) to the car's OBD-II port, on this board so the unterminated branch off the car's bus is only the ~0.5 m OBD lead. R59 holds TX recessive and R60 holds Rs high (standby, cannot transmit) whenever the pod is not driving them. No termination: the car has its two.




Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Under-dash power board (power, actuator, relays, CAN)
**AR Windscreen FYP — Kiogora Ian Mwenda**

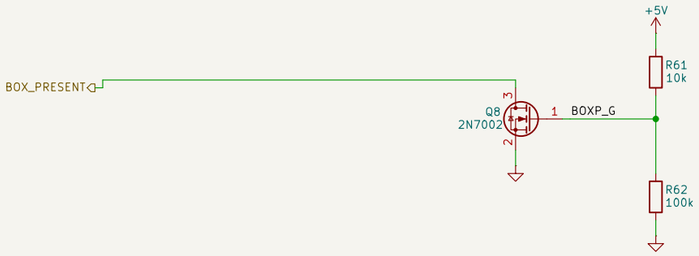
Sheet: /CAN transceiver for OBD-II/
File: can.kicad_sch

***Title: CAN transceiver for OBD-II***

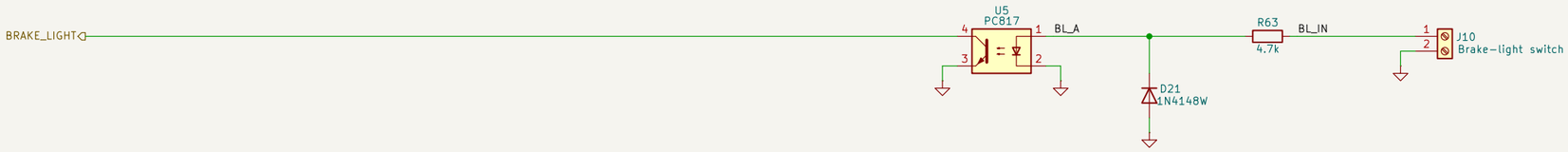
| Size: A4 | Date: 2026-10-07 | **Rev: 1** |
|---|---|---|
| KiCad E.D.A. 10.0.1 | | Id: 7/9 |

Inputs that report to the pod. Car side on the right, pod side (the ports) on the left.

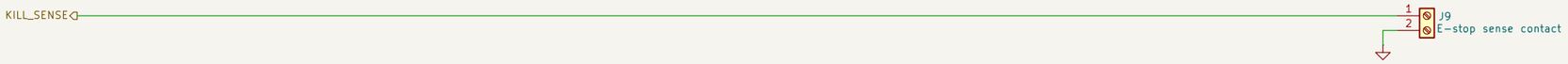
BOX_PRESENT: Q8 pulls it low only while this board's 5 V is up,
so the pod sees 'present' only when the box is connected AND powered.




Brake light: the opto-coupler U5 isolates the car's 12 V brake-light
circuit; D21 protects its LED from reverse voltage.



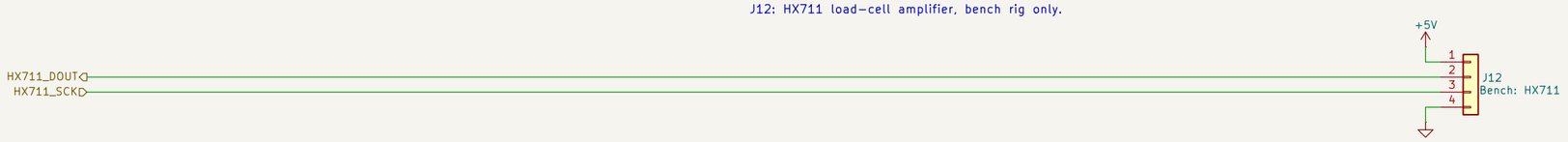

E-stop second contact (NC): released pulls KILL_SENSE low.




J8: +12V -> E-stop NC contact -> 40 A kill relay coil -> GND (wired off the board).




J12: HX711 load-cell amplifier, bench rig only.




Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Under-dash power board (power, actuator, relays, CAN)
**AR Windscreen FYP — Kiogora Ian Mwenda**

Sheet: /Presence, brake light, E-stop sense, kill relay, bench/
File: status.kicad_sch

***Title: Presence, brake light, E-stop sense, kill relay, bench***

| Size: A3 | Date: 2026-10-07 | **Rev: 1** |
|---|---|---|
| KiCad E.D.A. 10.0.1 | | Id: 8/9 |

INTER-BOX CABLE (BUILD_GUIDE 4.8.3). Shield bonded to GND at this end only.
Every line driven by the pod has its fail-safe pull-down on the sheet that uses it.




Generated by hardware/gen; see BUILD_GUIDE Part 4.8
Under-dash power board (power, actuator, relays, CAN)
**AR Windscreen FYP — Kiogora Ian Mwenda**

Sheet: /Inter-box cable: DB-25 to the pod/
File: cable.kicad_sch

***Title: Inter-box cable: DB-25 to the pod***

| Size: A4 | Date: 2026-10-07 | **Rev: 1** |
|---|---|---|
| KiCad E.D.A. 10.0.1 | | Id: 9/9 |